



## PROTOCOL

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1. For each reaction you set up, **top-label a PCR tube** with the name(s) indicated on the labsheet.
2. Retrieve **water** (white **W** tubes) and **T4 ligase buffer** aliquots (**red** tubes) from the **enzyme freezer** and thaw them at room temperature.
3. Set up the reaction, in order:
  - 6 water
  - 1 10× T4 DNA ligase buffer
  - 2 DNA
  - 0.5 T4 DNA ligase
  - 0.5 restriction enzyme

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**10 µL total**

  - If you have **more than 2 fragments** to join, premix **equal volumes of each DNA** in a tube, then use **2 µL of that mix** for the reaction.
  - Be sure you are using the **right one** of BsaI, BsmBI, BseRI, AarI, SapI, BbsI, as indicated in your **construction file and labsheets**.
4. Retrieve the **enzyme cooler**, and add **0.5 µL each** of the ligase and the restriction enzyme to each sample.
5. **Cap, mix, spin.**
6. Run the **GG1** program on the thermocycler.